1 A car of mass 1800 kg is moving in a straight line. Fig. 1.1 shows the variation with time t of the momentum p of the car.

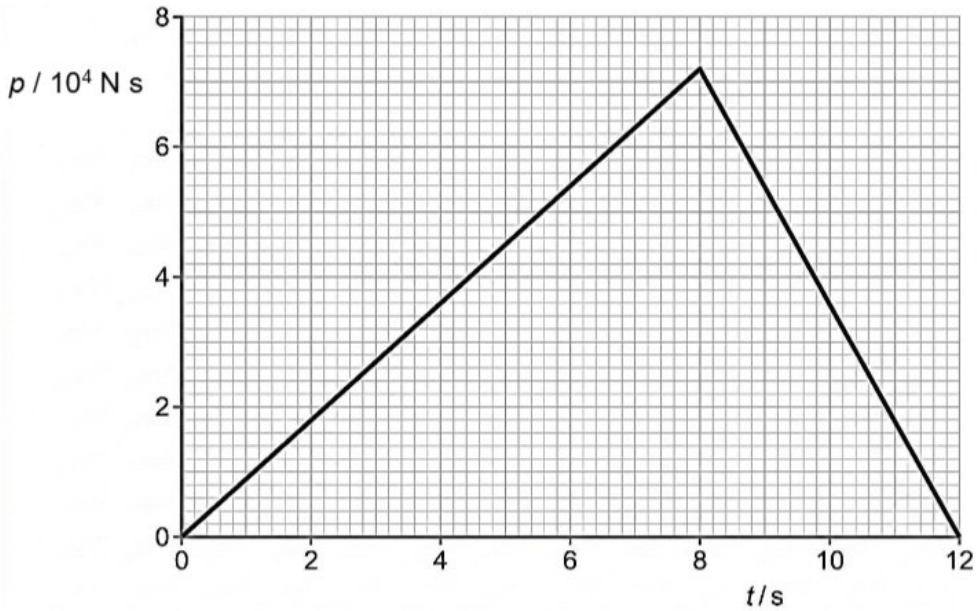


Fig. 1.1

(a) Calculate the maximum kinetic energy of the car.

kinetic energy = ... J [2]

(b) Determine the distance travelled by the car between times $t = 0$ and $t = 12$ s.

distance = ... m [2]

(c) On Fig. 1.2, sketch the variation with time t of the acceleration a of the car from $t = 0$ and $t = 12$ s.

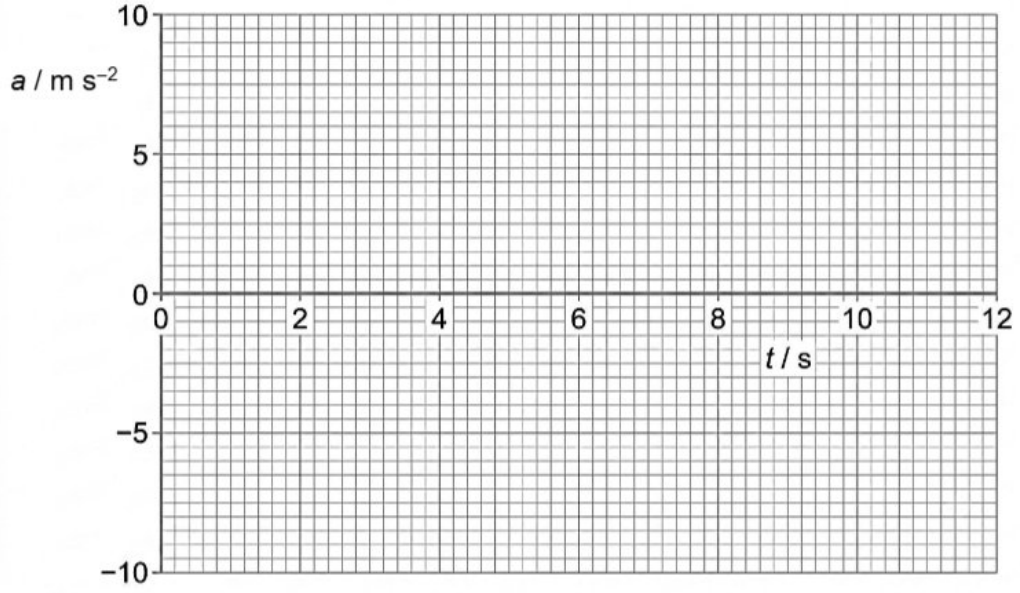


Fig. 1.2

[2]

[Total: 6]